

The antiaging approach for the treatment of dry eye.

Dry eye is one of the most common eye disorders affecting millions of people. It causes ocular irritation or discomfort, and decreases functional vision, causing a dramatic deterioration in the quality of life. Although new treatments such as the P2Y₂ agonist or cyclosporine eye drops have been developed and a certain level of patient satisfaction can now be obtained, no fundamental treatment has been developed. Currently, there is no therapy available to recover lacrimal function to its normal status. Recent progress in the understanding of aging has laid the foundations for a new way of thinking about intervention of the aging process. Because dry eye is accelerated by aging, a useful approach for the prevention or treatment of dry eye may be to interfere with the aging process. In the scientific community, there is a global consensus that calorie restriction can extend the life span of various kinds of animals, establishing an intervention to aging. Another important hypothesis believed to be involved in aging is the free radical theory. According to these theories, the aging process may be managed by controlling levels of calories or reactive oxygen species. In this review, these 2 important aging theories, calorie restriction and free radical aging, are examined, and we discuss how to apply these theories to the prevention and treatment of dry eye.